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Combining Forecasts from Multiple Experts for Multiple Variables

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Abstract. We propose a new method for combining point forecasts from multiple experts across multiple variables to improve inference about an unknown variable. Traditional aggregation methods combine forecasts for each variable separately (we refer to these methods as separate inference methods). However, in practice, each expert frequently provides a forecast for each variable across multiple variables. Because forecast errors can be correlated across the variables, the decision maker may additionally benefit from pooling forecasts for the other variables (we refer to our proposed method as pooled inference). Our model assumes that the forecast error arises from multiple sources of randomness, including a variable-specific factor, an expert-specific factor, a common factor, and idiosyncratic noise. When the covariance structure of the forecast errors is known, we show that the pooled inference is structurally equivalent to the separate inference but with reduced idiosyncratic noise, thereby allowing for more effective weighting of experts' forecasts. When the covariance structure is unknown, we derive the optimal form for the consensus forecast under the pooled inference, which can be easily implemented. Our empirical analysis demonstrates that the pooled inference can outperform widely used separate inference methods in some forecasting environments. Additionally, even when pooling a large number of variables—necessitating the handling of high-dimensional covariance matrices—the pooled inference remains computationally manageable.

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1. Introduction

Uncertainties are ubiquitous in many real-world decision-making contexts. For example, managers might be uncertain about the future demand for new products, which affects their inventory decisions, or policymakers might be uncertain about future macroeconomic conditions, such as the inflation rate or unemployment rate, which affect their policy decisions. To resolve these uncertainties, decision makers often obtain information in the form of point forecasts concerning an unknown variable from multiple information sources, which can be experts or quantitative models (Winkler 1981). For ease of exposition, we will refer to the information sources as experts. Over the past several decades, a significant amount of research has focused on aggregating forecasts from multiple experts about a single variable into a consensus forecast (see review articles by, e.g., Clemen 1989, Armstrong 2001, Petropoulos et al. 2022, and Wang et al. 2023). A well-known finding is that the consensus forecast, popularly known as "the wisdom of crowds," often demonstrates superior accuracy, sometimes even surpassing the most accurate individual forecast (Surowiecki 2005; Larrick and Soll 2006; Davis-Stober et al. 2014, 2015).

In practice, each expert may provide a forecast for each variable across multiple variables (Satopää 2021, Satopää et al. 2021). For example, in the retail industry, experts forecast sales of multiple products (Hammond and Raman 1994, Guo et al. 2025). In the financial industry, analysts cover multiple stocks (Palley et al. 2025), and economists forecast multiple macroeconomic indicators (Jose et al. 2014). In the agricultural industry, agricultural experts forecast the yields of multiple crops (Bansal and Gutierrez 2020). To draw an inference about an unknown variable, it is common to use a *separate inference*, in which the consensus forecast is based

only on the forecasts for that variable of interest (see, e.g., Bates and Granger 1969, Winkler 1981, and Soule et al. 2024). However, when forecasting multiple variables, the forecast errors may be correlated across the variables. Hence, compared with a separate inference, decision makers may additionally benefit from pooling experts' forecasts of other variables in order to draw an inference about the variable of interest (we refer to this as the *pooled inference*). This is in line with the notion that "inferences or decisions in the face of uncertainty should be based on all available information" (Winkler 1981, p. 479) as more information will lead to a more precise estimate regarding the unknown variable and is, therefore, more effective in reducing uncertainty (Clemen and Winkler 1985).

Motivated by the settings described above and by the potential informational value of pooling, we ask the following question. *When drawing an inference about a variable of interest, how can a decision maker incorporate forecasts of other variables in addition to those provided for the focal variable*?

To address the research question, we build on the model developed in Winkler (1981), which aggregates forecasts from multiple experts about a single variable. Because of their common training and experience, some dependence among experts' forecasts is inevitable (Broomell and Budescu 2009). In Winkler (1981), the dependence among experts is modeled via the dependence among their forecast errors. In addition to the correlated forecast errors with respect to a given variable (corresponding to the single-variable setting in Winkler 1981), forecast errors can also be correlated across different variables. To model such a covariance structure, we assume that forecast errors consist of four sources of randomness: (i) an error due to a variable-specific factor, (ii) an error due to an expert-specific factor, (iii) a common factor that drives all forecast errors, and (iv) idiosyncratic noise. The randomness in (i) and (ii) induces the forecast error correlation within and across variables, respectively; and the randomness in (iii) induces the correlation among all forecast errors. In particular, the randomness in (ii), which captures the crossvariable forecast error correlation, is the most crucial feature in our model as it sets the pooled inference apart from the separate inference.

1.1. Summary of Main Findings

Below, we provide a summary of our main findings. First, under a known forecast error covariance structure, we derive the posterior means (which represent the consensus forecast) and posterior variances under the pooled inference in closed form. The posterior mean is equal to the simple average of the forecasts for the focal variable plus an adjustment term. This adjustment term accounts for expert heterogeneity and is influenced by the forecasts of both the focal variable and other variables. To further understand this, we show that the adjustment term under the pooled inference is structurally equivalent to that under the separate inference but with smaller idiosyncratic noise. This reduced noise allows the decision maker to weigh the forecasts from the heterogeneous experts more judiciously under the pooled inference.

Despite this benefit, however, we find that there are two special cases where the value of additional information from pooling is zero. (i) The experts are exchangeable, or (ii) there is no idiosyncratic noise. That is, the pooled and separate inferences yield the same posterior distributions if either of the two conditions holds. To understand this, note that in (i), because of the exchangeability of the experts, there is no need to weigh their forecasts differently. Under the pooled inference, the consensus forecast, in this case, is equal to the simple average of the forecasts provided for the focal variable, exactly the same as it would be under the separate inference. In (ii), there is no further room to reduce the idiosyncratic noise; therefore, the noise reduction benefit under the pooled inference is muted, rendering the pooled and separate inferences equivalent. From a managerial perspective, a key takeaway is that a decision maker should only consider using the pooled inference when experts are substantially heterogeneous.

Finally, when the covariance structure of the forecast errors is unknown, we derive the optimal form or the consensus forecast and illustrate how it can be implemented using the linear shrinkage covariance estimator. Intuitively, the trade-off associated with the pooled inference lies between the informational gain and the increased estimation burden. Using submissions from the M4 and M5 competitions (Makridakis et al. 2020, 2022a), we demonstrate that pooled inference can outperform popular separate inference methods in some settings. Additionally, even when pooling a large number of variables—necessitating the handling of high-dimensional covariance matrices—the pooled inference remains computationally manageable. For example, in our simulation studies, when aggregating 15 forecasts across 7,000 variables, requiring the estimation of a $105{,}000 \times 105{,}000$ covariance matrix, the pooled inference performs well with a sample size of just 47 and takes less than 90 seconds to run.

1.2. Related Literature

In this paper, we address the pooled inference problem, in which a decision maker draws an inference about an unknown variable by pooling experts' forecasts for other variables in addition to the forecasts provided for the focal variable. This focus links our work to the broad literature on transfer and multitask learning. The former learns a new task by transferring knowledge from other related tasks (Pan and Yang 2010, Zhuang et al. 2021), whereas the latter solves

multiple learning tasks simultaneously by exploiting commonalities and differences across tasks (Zhang and Yang 2021). However, in both of these bodies of work, there is only one learning solution for each task. Translating this to our context, it would mean that there is only a single expert. Therefore, this literature does not speak to our research setting of forecast aggregation, where each unknown variable is often forecast by multiple experts. Hence, in the rest of this literature review, we mainly discuss prior work on forecast aggregation.

Combining multiple point forecasts into a consensus forecast has been shown to be an effective way to harness the wisdom of the crowd (Clemen 1989, Armstrong 2001, Wang et al. 2023). The most popular approach to forecast aggregation is the linear opinion pool, under which the consensus forecast is a linear combination of individual point forecasts (Stone 1961). A key step under this approach is determining the appropriate weights to assign to individual forecasts, which are based on the full knowledge of the covariance matrix of experts' forecast errors (Bates and Granger 1969, Winkler 1981). With limited data, it is challenging to estimate the covariance matrix, especially when the number of experts is large. Indeed, the optimal weighted aggregation using the sample covariance matrix cannot beat the simple equally weighted aggregation in many instances (Winkler and Makridakis 1983, Winkler 2015). The equally weighted aggregation treats experts as exchangeable. There is no need to differentiate between experts, and this model is, therefore, free from estimation errors. To strike a balance between a more complicated model and the associated estimation burden (also known as the bias-variance trade-off), other models have been proposed that only account for the heterogeneous variances, assuming independent experts (Bunn 1985, Schmittlein et al. 1990), or that account for the heterogeneous variances and a common pair-wise correlation among the experts (Soule et al. 2024). As another way of striking the bias-variance trade-off, Blanc and Setzer (2020) optimally combines the forecasts coming from an equally weighted aggregation and the optimally weighted aggregation using the sample covariance matrix derived based on Winkler (1981). In addition to the linear opinion pool, other combination methods have also been proposed, such as excluding extreme forecasts before aggregation (Jose and Winkler 2008) or selecting a subset of the crowd based on their accuracy and diversity (Mannes et al. 2014) or contribution weights (Budescu and Chen 2015). Despite the extensive literature, it is typically assumed in the above papers that to draw an inference about an unknown variable, the decision maker only uses forecasts provided for that variable of interest, which is equivalent to the case of the separate inference.

A separate stream of work has recently examined how to leverage additional information to aggregate forecasts. For example, Palley and Soll (2019) and Palley and Satopää (2023) study elicitation of both experts' own forecasts and their estimates of the average forecasts provided by their peers and then, using both of these in the aggregation. Bansal and Gutierrez (2020) calibrate forecasts of each expert using a standardized z-score, assuming that such standardized z-scores by the same expert are exchangeable across different variables. In a hierarchical context, Nystrup et al. (2020) investigate how to leverage information across different aggregation levels to reconcile forecasts. Empirically, using the linear shrinkage covariance estimator proposed by Ledoit and Wolf (2004), they demonstrate that there are significant benefits in capturing dependencies between forecast errors across different aggregation levels. In our paper, we also use the linear shrinkage covariance estimator to implement the pooled inference and demonstrate that it can perform well. However, the key difference is that their main objective is to reconcile forecasts, ensuring that the forecasts are coherent across different aggregation levels, whereas we focus on forecast aggregation.

To the best of our knowledge, prior literature that studies the pooled inference problem is scarce, with the exceptions of Agnew (1985) and Satopää (2021). At a high level, Satopää (2021, p. 1729) argues that the benefit of leveraging forecasts from other variables is that "the base outcome and experts' information structure can be estimated by 'borrowing strength' across outcomes." The pooled inference in our paper also has the flavor of "borrowing strength" (Satopää 2021, p. 1729). However, the channels through which the "borrowing strength" (Satopää 2021, p. 1729) occurs differ. Satopää (2021) assumes that the base outcomes for all variables are identical (i.e., they have the same unconditional expectation). This feature allows for "borrowing strength" across variables in Satopää (2021). In contrast, our paper does not require this assumption. In our paper, the key feature that allows for "borrowing strength" under the pooled inference is the correlation of forecast errors across variables (Satopää 2021, p. 1729).[1]

Agnew (1985), in contrast, analyzes the pooled inference problem by accounting for the forecast error correlation across variables. Our paper departs from Agnew (1985) in several important ways. First, Agnew (1985) does not offer any insights about comparing the pooled and separate inferences or about when pooling provides (does not provide) value in terms of additional information. Second, under the model of Agnew (1985), it can be shown that pooled inference is always equivalent to separate inference when the covariance matrix is known; see Online Appendix E.4 for the

proof. In other words, the value of additional information from pooling in Agnew (1985) is always zero. Third, up to a linear transformation, all of the cases considered in Agnew (1985) are equivalent to the cases where forecast errors are identically and independently distributed (i.i.d.) either across variables or across experts. However, our model goes beyond such i.i.d. cases. As such, our methodology could handle the cases considered in Agnew (1985) (see Online Appendix E.4), whereas the reverse is not true.

1.3. Structure of This Paper

The rest of this paper is organized as follows. First, in Section 2, we describe the model setup. Next, in Section 3, under a known forecast error covariance structure, we present the posterior mean and variance under the pooled inference and compare them with those under the separate inference. Subsequently, in Section 4, when the covariance structure is unknown, we derive the optimal form for the consensus forecast under the pooled inference. Lastly, in Section 5, we conduct an empirical analysis using submissions from the M4 and M5 competitions. Throughout the paper, we follow the notational conventions in Table 1.

2. Model

Suppose that there are n experts and m unknown variables. Expert j, $j = 1, \ldots, n$ provides a point forecast $f_{i,j}$ for variable i, $i = 1, \ldots, m$, whose true value is θ_i. Hence, the forecast error is given by $u_{i,j} = f_{i,j} - \theta_i$. Following Winkler (1981), we model dependence via the dependence among the forecast errors $u_{i,j}$. Analyzing a general covariance matrix offers little tractability and few insights (Clemen and Winkler 1985). Therefore, some structure is essential. Specifically, we assume that the forecast errors $u_{i,j}$ are due to four independent sources of randomness:

$$u_{i,j} = x_i + c_i(y_j + z) + \epsilon_{i,j}. \tag{1}$$

Below, we explain each source of randomness. First, x_i is the error due to the variable-specific factor, which induces dependence across experts for variable i. For ease of exposition, we normalize $Var(x_i) = 1$ for all variables $i = 1, \ldots, m$.[2] Second, y_j is the error due to the expert-specific factor, which induces dependence across variables for expert j. This across-variable dependence is critical to the pooled inference. Otherwise, if forecast errors across different variables are independent, the pooled inference is equivalent to the separate inference (see Lemma EC.2 in Online Appendix E, which formalizes this intuition). We allow y_j to be different for each expert j, with $Var(y_j) = \sigma_j^2$ $(j = 1, \ldots, n)$. The parameter σ_j represents the expert j's forecasting skill, with higher values of σ_j indicating noisier or worse experts. In other words, the expert heterogeneity in our model is represented through the heterogeneous variances of y_j. Third, z, with $Var(z) = \sigma_z^2$, represents the common factor that drives all of the forecast errors for all variables and experts. As a result, even if two forecast errors $u_{i,j}$ and $u_{i',j'}$ correspond to different experts and variables (i.e., $i \neq i'$ and $j \neq j'$), they are still correlated because of the common factor z. Fourth, $\epsilon_{i,j}$ is the idiosyncratic noise with $Var(\epsilon_{ij}) = \sigma_\epsilon^2$, independently and identically distributed for each pair i, j. We also assume that for all i and j, x_i, y_j, z, and $\epsilon_{i,j}$ are independent. In addition to the four sources of randomness, we incorporate constants c_i, $1 \leq i \leq m$, which are known and enable the possibility of negative correlations between forecast errors across different variables. For example, if we set (c_1, c_2) to $(1, -1)$ with $Var(z) = 0$, then $Cov(u_{1,j}, u_{2,j}) = -\sigma_j^2$, which indicates a negative correlation between $u_{1,j}$ and $u_{2,j}$.

The additive structure in (1) is commonly used in modeling dependence (Broomell and Budescu 2009), and more broadly, such decomposition of forecasts into various independent sources has also been adopted by Petropoulos et al. (2018). Following Winkler (1981), the four sources of randomness are assumed to be normally distributed, and each has a mean of zero. This normality assumption is also motivated by the "normal theory of errors," according to which normality provides a reasonable approximation to model forecast errors (Winkler 1981). The zero-mean assumption ensures that experts are unbiased. If an expert is biased, the decision maker can debias the expert using historical forecast errors.

Based on the data-generating process in (1), we can write the full covariance matrix $\Sigma_{\boldsymbol{u}}$ among all $m \times n$ forecast errors. For an explicit expression of the general $(mn) \times (mn)$ covariance matrix $\Sigma_{\boldsymbol{u}}$, we refer interested readers to Equation (EC.2) in Online Appendix A. To build intuition, we present the following illustrative example, which we will revisit throughout the paper to complement our general findings.

Table 1. Notational Conventions Used in This Paper

Font	Notation	Example
Capital letter	Matrix	$\Sigma_{\boldsymbol{u}_i}$: covariance matrix for the vector $\boldsymbol{u}_i$
Boldface	Vector	$\boldsymbol{u}_i = (u_{i,1}, \ldots, u_{i,n})^T$: length-$n$ column vector
Normal font	Scalar	$u_{i,j}$: the jth element of the vector $\boldsymbol{u}_i$

2.1. An Illustrative Example

To illustrate, we set $c_i = 1$ for $i = 1, 2, \ldots, m$ and assume $\sigma_z^2 = 0$. This simplification reduces the general data-generating process in (1) to $u_{i,j} = x_i + y_j + \epsilon_{i,j}$. Additionally, we consider a scenario with three variables $(m = 3)$ and two experts $(n = 2)$. For this simplified

scenario, Equation (2) below provides the 6×6 covariance matrix $\Sigma_{\boldsymbol{u}}$ for $\boldsymbol{u} = (u_{1,1}, u_{1,2}, u_{2,1}, u_{2,2}, u_{3,1}, u_{3,2})^T$:

$$\Sigma_{\boldsymbol{u}} = \begin{pmatrix} 1+\sigma_1^2+\sigma_\epsilon^2 & 1 & \sigma_1^2 & 0 & \sigma_1^2 & 0 \\ 1 & 1+\sigma_2^2+\sigma_\epsilon^2 & 0 & \sigma_2^2 & 0 & \sigma_2^2 \\ \sigma_1^2 & 0 & 1+\sigma_1^2+\sigma_\epsilon^2 & 1 & \sigma_1^2 & 0 \\ 0 & \sigma_2^2 & 1 & 1+\sigma_2^2+\sigma_\epsilon^2 & 0 & \sigma_2^2 \\ \sigma_1^2 & 0 & \sigma_1^2 & 0 & 1+\sigma_1^2+\sigma_\epsilon^2 & 1 \\ 0 & \sigma_2^2 & 0 & \sigma_2^2 & 1 & 1+\sigma_2^2+\sigma_\epsilon^2 \end{pmatrix}. \tag{2}$$

Using the covariance information in (2), the decision maker aims to draw an inference about each of the individual unknown variables θ_i $(i = 1, 2, 3)$ by aggregating forecasts. For separate inference, the decision maker only uses $\Sigma_{\boldsymbol{u}_i} = \begin{pmatrix} 1+\sigma_1^2+\sigma_\epsilon^2 & 1 \\ 1 & 1+\sigma_2^2+\sigma_\epsilon^2 \end{pmatrix}$, the block diagonal of $\Sigma_{\boldsymbol{u}}$, to aggregate forecasts $f_{i,1}$ and $f_{i,2}$ provided for θ_i (Winkler 1981). In contrast, for pooled inference, the decision maker uses $\Sigma_{\boldsymbol{u}}$ to aggregate all of the available forecasts $\boldsymbol{f} = (f_{1,1}, f_{1,2}, f_{2,1}, f_{2,2}, f_{3,1}, f_{3,2})^T$. For derivation details of the separate and pooled inferences, we refer interested readers to Online Appendices B and C, respectively.

Throughout our analysis, we use the diffuse (flat) prior, which represents the situation in which the decision maker has little information or a weak prior about the unknown variables. For this reason, our results—specifically, any difference in the posterior distributions between the pooled and separate inferences—can only be attributed to the nature of the inference methods and not the choice of priors (see, e.g., Gaba et al. 2019, Satopää et al. 2021, and Palley and Satopää 2023 for similar treatments of the priors).

3. Main Theoretical Results Under a Known Covariance Matrix

In this section, we present the results under the general data-generating process outlined in (1) when there are m variables and n experts. We start with posterior means (Section 3.1) followed by posterior variances (Section 3.2). Throughout this section, we often complement the general findings with the illustrative example presented in Section 2.1 in order to convey insights.

3.1. Posterior Means

We first present the posterior means under the separate and pooled inferences (Proposition 1) and then compare the two (Corollary 1).

Proposition 1. *Suppose that there are n experts and m variables and that each expert $j = 1, \ldots, n$ provides a point forecast $f_{i,j}$ for the variable $i = 1, \ldots, m$.*

i. *Under the separate inference, the posterior mean $\mu^\star_{\theta_i,s}$ of θ_i is*

$$\mu^\star_{\theta_i,s} = \frac{1}{n}\sum_{j=1}^{n} f_{i,j} + \sum_{j=1}^{n-1} l_{j,s}(f_{i,j} - f_{i,n}), \text{ where}$$

$$l_{j,s} = \frac{\frac{1}{\sigma_j^2+\sigma_\epsilon^2/c_i^2}}{\sum_{j=1}^{n}\frac{1}{\sigma_j^2+\sigma_\epsilon^2/c_i^2}} - \frac{1}{n}. \tag{3}$$

ii. *Under the pooled inference, the posterior mean $\mu^\star_{\theta_i,p}$ of θ_i is*

$$\mu^\star_{\theta_i,p} = \frac{1}{n}\sum_{j=1}^{n} f_{i,j} + \sum_{j=1}^{n-1} l_{j,p}\left(\sum_{k=1}^{m}\frac{c_i c_k}{\|\boldsymbol{c}\|_2^2}(f_{k,j} - f_{k,n})\right), \text{ where}$$

$$l_{j,p} = \frac{\frac{1}{\sigma_j^2+\sigma_\epsilon^2/\|\boldsymbol{c}\|_2^2}}{\sum_{j=1}^{n}\frac{1}{\sigma_j^2+\sigma_\epsilon^2/\|\boldsymbol{c}\|_2^2}} - \frac{1}{n}, \tag{4}$$

where $\|\boldsymbol{c}\|_2$ denotes the l_2-norm. Therefore, $\|\boldsymbol{c}\|_2^2 = \sum_{i=1}^{n} c_i^2$.

Parts (i) and (ii) of Proposition 1 present the posterior means of θ_i $(i = 1, 2, \ldots, m)$ under the separate and pooled inferences, respectively. Both are equal to the sum of two parts: an equally weighted aggregation and an adjustment term. The equally weighted aggregation is the same in both (3) and (4), taking a simple average of all forecasts provided for the focal variable. For the adjustment term, without the loss of generality, we use the forecasts of the nth expert as the reference forecast, and the adjustment is calculated based on how forecasts from the other experts (i.e., expert $j = 1, \ldots, n-1$) deviate from the reference forecast. Despite the similarity, the adjustment terms are different in (3) and (4). Under the separate inference, the adjustment term in (3) only uses the forecasts for the variable θ_i. In contrast, under the pooled inference, the adjustment term in (4) uses all of the available forecasts.

To gain a more intuitive understanding of the adjustment terms under the separate and pooled inferences, we revisit our illustrative example in Section 2.1 by setting $c_i = 1$ and $\sigma_z^2 = 0$ under (1), with three variables $(m = 3)$ and two experts $(n = 2)$. The covariance matrix for this scenario is provided in (2). Under this special scenario, the posterior means for $\theta_i, i = 1, 2, 3$, can be written as (note that because $n = 2$, for the ease of reading, we omit the subscript j from $l_{j,s}$ and $l_{j,p}$ and directly present l_s and l_p in the following two expressions)

$$\mu^\star_{\theta_i,s} = \frac{1}{2}(f_{i,1} + f_{i,2}) + l_s(f_{i,1} - f_{i,2}), \text{ where}$$

$$l_s = \frac{\sigma_2^2 - \sigma_1^2}{2\sigma_1^2 + 2\sigma_2^2 + 4\sigma_\epsilon^2}. \tag{5}$$

$$\mu^\star_{\theta_i,p} = \frac{1}{2}(f_{i,1} + f_{i,2}) + l_p\frac{\sum_{k=1}^{3}(f_{k,1} - f_{k,2})}{3}, \text{ where}$$

$$l_p = \frac{\sigma_2^2 - \sigma_1^2}{2\sigma_1^2 + 2\sigma_2^2 + 4\sigma_\epsilon^2/3}. \tag{6}$$

Under the separate inference, the adjustment term $l_s(f_{i,1}-f_{i,2})$ in (5) is straightforward to interpret; it adjusts the equally weighted aggregation toward the less noisy expert with lower σ_j^2 (the sign of l_s determines this direction of adjustment). Does this intuition carry over to the pooled inference? To answer this question, we look at the two individual parts of the adjustment term in (6): (a) the forecast difference(s) $\sum_{k=1}^{3}(f_{k,1}-f_{k,2})/3$ and (b) the coefficient l_p. We also contrast these terms with those under the separate inference.

To understand (a), let us go back to the data-generating process under the illustrative example: $f_{i,j}=\theta_i+x_i+y_j+\epsilon_{i,j}$. Under the pooled inference, the forecast differences in (6) are given by $\sum_{k=1}^{3}(f_{k,1}-f_{k,2})/3=(y_1-y_2)+(\epsilon_{i,1}^{new}-\epsilon_{i,2}^{new})$, where $\epsilon_{i,1}^{new}=1/3\sum_{k=1}^{3}\epsilon_{k,1}$ and $\epsilon_{i,2}^{new}=1/3\sum_{k=1}^{3}\epsilon_{k,2}$. In contrast, the forecast difference in (5) is given by $f_{i,1}-f_{i,2}=(y_1-y_2)+(\epsilon_{i,1}-\epsilon_{i,2})$. Comparing the two, it is as if the idiosyncratic noise under the pooled inference becomes $\epsilon_{i,1}^{new}$ and $\epsilon_{i,2}^{new}$, in contrast to $\epsilon_{i,1}$ and $\epsilon_{i,2}$ under the separate inference. It is easy to verify that the variances of the idiosyncratic noises satisfy the following relationship: $Var(\epsilon_{i,1}^{new})=Var(\epsilon_{i,2}^{new})=1/3Var(\epsilon_{i,1})=1/3Var(\epsilon_{i,2})=\sigma_\epsilon^2/3$, which implies that the pooled inference helps reduce the variability of the idiosyncratic noise. Put differently, the pooled and separate inferences are structurally the same, with the pooled inference having a smaller idiosyncratic noise. This structural equivalence between the pooled and separate inferences also carries over to the comparison between the coefficients l_p and l_s; we obtain l_p by replacing σ_ϵ^2 in l_s (in (5)) with $\sigma_\epsilon^2/3$. The sign of l_p determines the direction of the adjustment, just like the l_s in the separate inference. Therefore, under the pooled inference, the adjustment term $l_p\sum_{k=1}^{3}(f_{k,1}-f_{k,2})/3$ in (6) also helps to adjust the equally weighted aggregation toward the forecast of the less noisy expert. It is worth noting that the preceding discussion on the structural equivalence between the pooled and separate inferences applies to the adjustment terms of the posterior means (i.e., the second term in (3) and (4)). The equivalence does not directly carry over to the posterior means and posterior variances. Establishing a similar kind of equivalence in terms of the posterior means and posterior variances requires further refinements to our arguments. We formally discuss such refinements in detail in Online Appendix E.2.

Based on our discussion so far, the posterior means under the pooled and separate inferences are generally different because they are based on different information. However, there are two special cases in which their posterior means are identical as we will show in the following corollary.

Corollary 1. *Suppose that there are n experts and m variables. Then, the posterior means of θ_i under the pooled and separate inferences are identical if either of the following conditions is true.* (i) *The experts are exchangeable* ($\sigma_1=\cdots=\sigma_n$), *or* (ii) *there is no idiosyncratic noise* ($\sigma_\epsilon=0$).

Corollary 1 identifies two situations where the posterior means are the same under the pooled and separate inferences. To understand them, we note that first, under (i), when the experts are exchangeable ($\sigma_1=\cdots=\sigma_n$), it is easy to verify that $l_{j,s}=l_{j,p}=0$ for any $j=1,\ldots,n$ based on (3) and (4). Hence, regardless of whether the inference is pooled or separate, the posterior means of θ_i are determined by an equally weighted aggregation $(1/n)\sum_{j=1}^{n}f_{i,j}$. Second, under (ii), if there is no idiosyncratic noise ($\sigma_\epsilon=0$), it is also easy to verify that (3) is equal to (4) because the respective terms in both expressions are equal: that is, $l_{j,s}=l_{j,p}=\left(\sum_{j=1}^{n}(\sigma_j^{-2})\right)^{-1}(\sigma_j^{-2})-1/n$ and $f_{i,j}-f_{i,n}=\sum_{k=1}^{m}\frac{c_ic_k}{\|c\|_2^2}(f_{k,j}-f_{k,n})=c_i(y_j-y_n)$ for $j=1,\ldots,n-1$.

To further grasp the intuition of Corollary 1, recall that the pooled inference derives its benefits from the reduced idiosyncratic noise so that the decision maker can weigh the forecasts from heterogeneous experts more judiciously. Therefore, when the experts are exchangeable (so there is no need to weigh forecasts differently) or when there is no idiosyncratic noise (so there is no further room to reduce the idiosyncratic noise), the posterior means are identical under the pooled and separate inferences.

3.1.1. An Alternative View of the Posterior Means. In our discussion above (e.g., see Proposition 1), we have parameterized the posterior means as the sum of an equally weighted aggregation and an adjustment term. To enrich our understanding, the posterior means can also be rewritten as a weighted average of forecasts. To gain more insights into these weights, we again revisit the illustrative example in Section 2.1 by setting $c_i=1$ and $\sigma_z^2=0$ under (1), with three variables ($m=3$) and two experts ($n=2$). Without the loss of generality, we focus on the variable θ_1. We can rearrange (5) and (6) as an weighted average of the forecasts ($f_{1,1}$,$f_{1,2}$,$f_{2,1}$, $f_{2,2}$,$f_{3,1}$,$f_{3,2}$). The weights on each of the forecasts are summarized in Table 2. In the subsequent discussion, we use superscripts to differentiate weights under the separate and pooled inferences. For example, $w_{1,1}^S$ ($w_{1,1}^P$) refers to the weight assigned to $f_{1,1}$ under the separate (pooled) inference.

The question of how to assign forecast weights under the separate inference is well understood in the literature (Winkler and Clemen 1992). As expected, to draw inference about θ_1, under the separate inference, only forecasts provided for θ_1 (i.e., $f_{1,1}$ and $f_{1,2}$) are used, and forecasts provided for θ_2 (i.e., $f_{2,1}$ and $f_{2,2}$)

Table 2. Summary of Forecast Weights Based on the Illustrative Example in Section 2.1 with Three Variables and Two Experts

	$w_{1,1}$	$w_{1,2}$	$w_{2,1} = w_{3,1}$	$w_{2,2} = w_{3,2}$	$w_{1,1} + w_{2,1} + w_{3,1}$	$w_{1,2} + w_{2,2} + w_{3,2}$
Separate	$\frac{1}{2} + \frac{\sigma_2^2 - \sigma_1^2}{2\sigma_1^2 + 2\sigma_2^2 + 4\sigma_\epsilon^2}$	$\frac{1}{2} - \frac{\sigma_2^2 - \sigma_1^2}{2\sigma_1^2 + 2\sigma_2^2 + 4\sigma_\epsilon^2}$	0	0	$\frac{1}{2} + \frac{\sigma_2^2 - \sigma_1^2}{2\sigma_1^2 + 2\sigma_2^2 + 4\sigma_\epsilon^2}$	$\frac{1}{2} - \frac{\sigma_2^2 - \sigma_1^2}{2\sigma_1^2 + 2\sigma_2^2 + 4\sigma_\epsilon^2}$
Pooled	$\frac{1}{2} + \frac{\sigma_2^2 - \sigma_1^2}{6\sigma_1^2 + 6\sigma_2^2 + 4\sigma_\epsilon^2}$	$\frac{1}{2} - \frac{\sigma_2^2 - \sigma_1^2}{6\sigma_1^2 + 6\sigma_2^2 + 4\sigma_\epsilon^2}$	$\frac{\sigma_2^2 - \sigma_1^2}{6\sigma_1^2 + 6\sigma_2^2 + 4\sigma_\epsilon^2}$	$-\frac{\sigma_2^2 - \sigma_1^2}{6\sigma_1^2 + 6\sigma_2^2 + 4\sigma_\epsilon^2}$	$\frac{1}{2} + \frac{\sigma_2^2 - \sigma_1^2}{2\sigma_1^2 + 2\sigma_2^2 + \frac{4}{3}\sigma_\epsilon^2}$	$\frac{1}{2} - \frac{\sigma_2^2 - \sigma_1^2}{2\sigma_1^2 + 2\sigma_2^2 + \frac{4}{3}\sigma_\epsilon^2}$

Notes. The decision maker aggregates forecasts to infer θ_1. It is worth noting that $w_{2,1}^P + w_{2,2}^P = 0$ (implying that either $w_{2,1}^P$ or $w_{2,2}^P$ is negative) and $w_{3,1}^P + w_{3,2}^P = 0$ (implying that either $w_{3,1}^P$ or $w_{3,2}^P$ is negative). Except for this, all other weights in this table are nonnegative.

and θ_3 (i.e., $f_{3,1}$ and $f_{3,2}$) are not used, therefore receiving zero weights. Let us further suppose that $\sigma_1 < \sigma_2$ (i.e., expert 1 is more skillful); according to Table 2, we have $w_{1,1}^S > 1/2$ and $w_{1,2}^S < 1/2$ (i.e., expert 1 receives higher weight than expert 2). Moreover, $w_{1,1}^S$ decreases in σ_ϵ^2 (i.e., the more skillful expert receives even higher weight when there is less idiosyncratic noise).

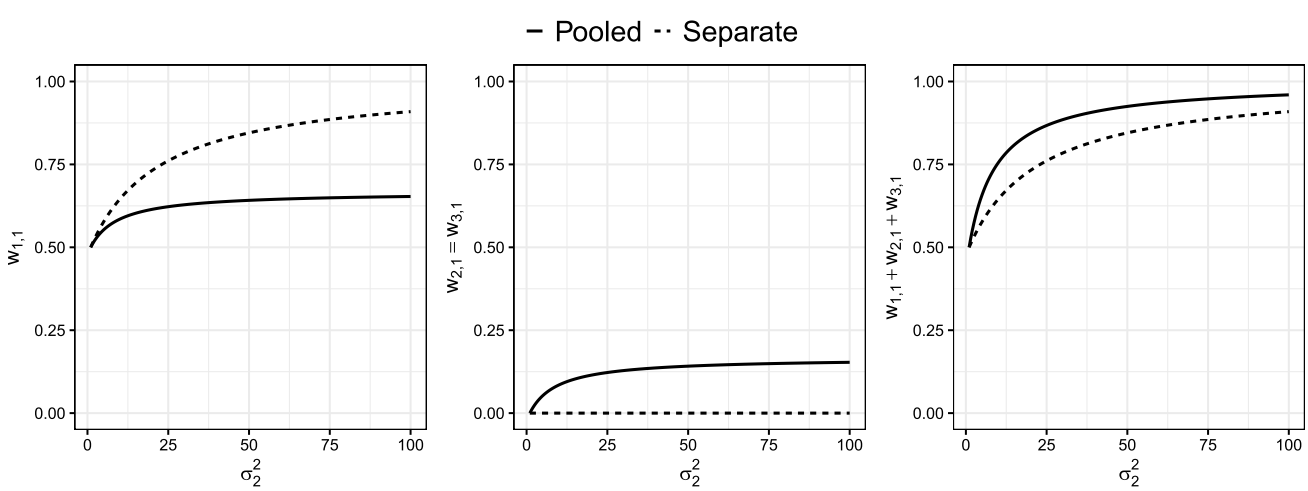
Because the pooled inference is structurally equivalent to the separate inference but with smaller idiosyncratic noise, one might naively expect that the weight assigned to $f_{1,1}$ under the pooled inference is higher than that under the separate inference (i.e., $w_{1,1}^P > w_{1,1}^S$). However, this intuition is faulty; in fact, the opposite is true (one can easily verify that $w_{1,1}^P < w_{1,1}^S$ from Table 2). To understand this seemingly counterintuitive result, it is instructive to realize that under the pooled inference, the weights assigned to expert 1 are reflected in $f_{1,1}$, $f_{2,1}$, and $f_{3,1}$. Hence, instead of focusing on $w_{1,1}^P$, it makes more sense to look at the overall weight assigned to expert 1, $w_{1,1}^P + w_{2,1}^P + w_{3,1}^P$, and compare it with $w_{1,1}^S + w_{2,1}^S + w_{3,1}^S$. Based on the second-to-last column in Table 2, it is evident that $w_{1,1}^P + w_{2,1}^P + w_{3,1}^P > w_{1,1}^S + w_{2,1}^S + w_{3,1}^S$ (i.e., the more skillful expert, expert 1, is assigned a higher overall weight under the pooled inference than that under the separate inference).

Figure 1 visualizes the weights under the separate and pooled inferences. In the simulation, we fix $\sigma_1^2 = 1$, $\sigma_\epsilon^2 = 10$ and vary σ_2^2 from 1 to 100. In this way, expert 1 is more skillful than expert 2. Overall, the patterns shown in Figure 1 are consistent with our discussion above. The left and center panels of Figure 1 show the individual weights assigned to expert 1's forecasts on θ_1: that is, $f_{1,1}$, $f_{2,1}$, and $f_{3,1}$, respectively. More specifically, in the left panel of Figure 1, $f_{1,1}$ receives a lower weight under the pooled inference than that under the separate inference $w_{1,1}^P < w_{1,1}^S$. In the center panel of Figure 1, $f_{2,1}$ and $f_{3,1}$ receive positive weights under the pooled inference and zero weights under the separate inference. The right panel of Figure 1 shows the overall weights assigned to expert 1; it is evident that expert 1 is assigned a higher overall weight under the pooled inference than under the separate inference. The above insight does not translate directly to weights in general when $c_i \neq 1$. However, it holds under a modified version of weights. We formally discuss this modification in Online Appendix E.3.

3.2. Posterior Variances

So far, we have compared the posterior means of θ_i under the pooled and separate inferences. But, how

Figure 1. Examples of Weights Under Three Variables and Two Experts



Notes. In this numerical example, the decision maker aggregates forecasts to infer about θ_1. The parameter values used are $\sigma_1^2 = 1$ and $\sigma_\epsilon^2 = 10$.

do the posterior variances of θ_i compare? Proposition 2 formally answers this question.

Proposition 2. *Suppose that there are n experts and m variables. If we denote the posterior variance of $\theta_i (i = 1, \ldots, m)$ as $\sigma^{\star 2}_{\theta_i,s}$ (respectively, $\sigma^{\star 2}_{\theta_i,p}$) under the separate inference (respectively, pooled inference), then for $i = 1, \ldots, m$, we have the following relationship between the posterior variances*:

$$\Delta_{\theta_i} = \sigma^{\star 2}_{\theta_i,s} - \sigma^{\star 2}_{\theta_i,p} \geq 0. \tag{7}$$

Moreover, $\Delta_{\theta_i} = 0$ if either of the following conditions is true. (i) *The experts are exchangeable* ($\sigma_1 = \cdots = \sigma_n$), *or* (ii) *there is no idiosyncratic error* ($\sigma_\epsilon = 0$).

Proposition 2 compares posterior variances of $\theta_i (i = 1, \ldots, m)$ between the pooled and separate inferences under the general case of n experts and m variables. Two important insights are immediately clear based on (7). First, the difference Δ_{θ_i} is nonnegative, which implies that the posterior variance under the separate inference is at least as large as that under the pooled inference. This result is intuitive because under the pooled inference, the decision maker uses more information, and the additional information leads to a more precise estimate of θ_i. Thus, the reduction in the posterior variance can also be seen as the value of the additional information from pooling. Second, the difference Δ_{θ_i} is not always strictly positive. It is equal to zero when the experts are exchangeable ($\sigma_1 = \cdots = \sigma_n$), or there is no idiosyncratic noise ($\sigma_\epsilon = 0$). This insight is consistent with Corollary 1. In either situation, it is as if the additional information from the forecasts for other variables is redundant. Therefore, there is no reduction in variance, or equivalently, the value of the additional information from pooling is zero.

We can further understand Δ_{θ_i} by revisiting the illustrative example in Section 2.1. In this case, we can explicitly express $\Delta_{\theta_i}, i = 1, 2,$ as

$$\Delta_{\theta_i} = \sigma^{\star 2}_{\theta_i,s} - \sigma^{\star 2}_{\theta_i,p} = \frac{(\sigma_2^2 - \sigma_1^2)^2 \sigma_\epsilon^2/3}{(\sigma_1^2 + \sigma_2^2 + 2\sigma_\epsilon^2)(\sigma_1^2 + \sigma_2^2 + 2\sigma_\epsilon^2/3)} \geq 0. \tag{8}$$

Because the variance of the variable-specific randomness x_i is normalized to be one or $Var(x_i) = 1$ for $i = 1, 2$ in our model, the difference Δ_{θ_i} shown in (8) is the same for $i = 1, 2$. If we fix the total variance $(\sigma_1^2 + \sigma_2^2)$ of the expert-specific randomness, the difference Δ_{θ_i} is increasing in the gap $(\sigma_1^2 - \sigma_2^2)$. In other words, there is a larger reduction in variance, or equivalently, the value of the additional information from pooling is higher when the experts become more heterogeneous in σ_j^2, $j = 1, 2$. In this case, there is a greater need to weigh their forecasts differently. The noise reduction benefit due to pooling helps fulfill that need more effectively.

In sum, the key takeaway here is that pooling offers no benefit when experts are exchangeable. This is likely to be the case when experts have similar skills (e.g., similar experiences) in judgmental forecasting or when experts use similar models in quantitative forecasting. This conclusion is derived under the assumption that the covariance matrix is known. Consequently, when the covariance matrix is unknown, pooling is expected to perform even worse than separate inference under the same condition of exchangeable experts. This is because under an unknown covariance structure, the pooled inference introduces greater estimation errors because of the need to estimate a larger covariance matrix compared with the separate inference. Therefore, in practical settings where the covariance matrix is unknown, it only makes sense to consider the pooled inference when experts are substantially heterogeneous. In addition, another implicit prerequisite for pooled inference is that decision makers should only pool related variables. Otherwise, when forecast errors across variables are independent, the inference is equivalent to the separate inference under a known covariance matrix. Bearing these insights in mind, we next proceed to study the pooled inference under an unknown covariance structure.

4. Pooled Inference Under an Unknown Covariance Matrix

How should the pooled inference be implemented when the covariance matrix is unknown but needs to be estimated from data? To start, it is important to note that empirically estimating the covariance matrix based on the data-generating process in (1) is not feasible as neither the true data-generating process nor the values of the various modeling primitives are known (see endnote 2). As a result, we do not impose any empirical restrictions on the covariance structure. Instead, we adopt the framework of an unknown covariance matrix in Winkler (1981, section 3.3), which recommends using an inverted Wishart distribution—a natural conjugate prior for the covariance matrix in a normal model—as a prior for the unknown $\Sigma_{\boldsymbol{u}}$. Based on this, the following proposition provides the posterior mean of $\boldsymbol{\theta}$ in closed form.

Proposition 3. *Suppose that $\boldsymbol{\theta}$ and $\Sigma_{\boldsymbol{u}}$ are independent a priori, with an improper diffuse density for $\boldsymbol{\theta}$ and an inverse Wishart density for $\Sigma_{\boldsymbol{u}}$ (with the degrees of freedom ν and scale matrix Σ_0^{-1}). Then, the posterior mean of $\boldsymbol{\theta}$ given $\boldsymbol{f}$ is*

$$(E^T \Sigma_0^{-1} E)^{-1} E^T \Sigma_0^{-1} \boldsymbol{f}, \tag{9}$$

where $E = I_{m\times m} \otimes \mathbf{1}_n$. Here, $I_{m\times m}$ is an $m \times m$ identity matrix, $\otimes$ is Kronecker product, and $\mathbf{1}_n$ is a length-n column vector.

As a special case of Proposition 3, if we set $m = 1$, the posterior mean in (9) is reduced to $(\mathbf{1}_n^T \Sigma_0^{-1} \boldsymbol{f})/(\mathbf{1}_n^T \Sigma_0^{-1} \mathbf{1}_n)$,

aligning with the separate inference result in Winkler (1981, equation 3.15) under an unknown covariance matrix. In general, when $m > 1$, Proposition 3 extends the result to the pooled inference. To implement (9), we need to know the covariance prior Σ_0, which can be determined based on subjective judgments and past data. In this paper, we follow Winkler (1981) and use past data to estimate Σ_0. In particular, we employ the linear shrinkage covariance estimator (Ledoit and Wolf 2004), which is a popular covariance estimation method that has demonstrated excellent performance in forecasting (Nystrup et al. 2020) as well as in a wide range of other domains, such as finance, neuroscience, and signal processing (Ledoit and Wolf 2022). This estimator shrinks the sample covariance matrix toward the identity matrix to achieve a good bias-variance tradeoff. As shown in Proposition 3, when the covariance matrix is the identity matrix, the pooled inference assigns equal weights to the forecasts of the focal variable. Therefore, the linear shrinkage covariance estimator helps shrink the pooled inference toward equally weighted aggregation. For implementation, we use the linshrink_cov function from the nlshrink R package (Ramprasad 2016) and denote this version of the pooled inference as P:Linear.

There are alternative covariance estimators. However, they are not suitable for the pooled inference. Among the popular ones, the sample covariance estimator (Winkler 1981), the robust covariance estimator (Rousseeuw and Leroy 1987, Rousseeuw and Driessen 1999), and the shrinkage estimator proposed by Blanc and Setzer (2020) all require the sample covariance matrix to be invertible. However, this condition is often not met under the pooled inference as the number of observations is typically smaller than the dimension (i.e., the number of rows or columns) of the covariance matrix. In addition, the variance weights estimator (Bunn 1985, Schmittlein et al. 1990) is also unsuitable because it estimates only the diagonal elements of the covariance matrix, setting all off-diagonal elements to zero. As a result, pooled inference is no different from separate inference (see Lemma EC.2 in Online Appendix E.1). Finally, the common correlation estimator (Soule et al. 2024) faces similar challenges as the variance weights estimator as the estimated common correlation is close to zero in our experiments.

5. Empirical Analysis

In this section, we seek to compare the empirical performance of the pooled inference method with benchmarks that belong to the separate inference methods. We conduct the empirical analysis using two data sets: submissions from the M4 and M5 competitions—two of the flagship competitions in forecasting (Makridakis et al. 2020, 2022a). The two competitions focus on different domains and feature the success of different quantitative forecasting methods, offering a rich environment to test the empirical performance of the pooled inference.

To streamline the presentation for the rest of this section, first, in Section 5.1, we introduce the M4 and M5 competition data sets. Next, in Section 5.2, we introduce popular benchmark methods that are based on separate inference. Subsequently, we describe the empirical procedures in Section 5.3. In Section 5.4, we conduct an empirical analysis to compare the performance of the pooled inference method with the benchmark methods on each of the two data sets. In Section 5.5, we perform additional simulations to substantiate the empirical findings. Finally, we close this section in Section 5.6 with a discussion.

5.1. Introducing the Data Sets

The M4 and M5 competitions held in 2018 and 2020, respectively, are the fourth and fifth in the series of M competitions, which have been the flagship competitions in forecasting since the first M competition in 1981. Over the years, the M competitions have attracted seasoned practitioners and academicians to test and evaluate the performance of forecasting methods on large-scale data sets. The competitions have contributed to advancing both the theory and the practice of forecasting (Taleb 2007, Hyndman 2020). Both M4 and M5 competitions involve point forecasts and probabilistic forecasts (e.g., intervals or quantiles). To align with the topic of the current paper, we are only concerned with the point forecast submissions.

Table 3 provides a high-level summary of the two data sets. Only 49 teams participated in the M4 competition, with 17 teams outperforming the best benchmark method used in the competition (i.e., theta benchmark) (see Makridakis et al. 2020, p. 61). The participating teams used a wide variety of methods, including pure statistical methods, hybrid methods, or aggregation of forecasts generated by these different methods (Makridakis et al. 2020). In contrast, 5,507 teams participated in the M5 competition, with 415 teams outperforming the best benchmark used in the competition (i.e., ES_bu) (see Makridakis et al. 2022a, p. 1352). The submissions of the top 50 teams were made available by the competition organizer. As documented by Makridakis et al. (2022a, p. 1346), most of them adopt the LightGBM model (Ke et al. 2017), a popular machine learning model widely used in forecasting competitions on Kaggle (Bojer and Meldgaard 2021).

Moreover, participants in the M5 competition were asked to forecast Walmart sales of various hierarchical levels (e.g., sales of a state, a store, a category, etc.). The hierarchical structure entails that different variables on the same level tend to be closely related.

Table 3. A High-Level Summary of the M4 and M5 Data Sets

	M4 competition	M5 competition
Methods used by best-performing teams	Statistical, hybrid, or combination	LightGBM
Total no. of participating teams	49	5,507
No. of teams outperforming benchmarks	17	415
Hierarchical	No	Yes

Notes. In the hybrid approach, seasonality and smoothing coefficients are fitted concurrently with the recurrent neural network weights by the same gradient descent (Makridakis et al. 2020), whereas combination refers to combining statistical and/or machine learning methods. LightGBM (Ke et al. 2017) is a popular machine learning method and is considered to be a standard method of choice in recent Kaggle competitions (Bojer and Meldgaard 2021).

Therefore, it makes the M5 data set a natural starting point to consider pooling the forecasts of the other variables on the same aggregation level. For this reason and for the ease of exposition, in the current and subsequent subsections, we organize the content in reverse chronological order; we start by presenting the M5 data set followed by the M4 data set.

5.1.1. M5 Data Description. The accuracy track of the M5 competition is dedicated to point forecasts. In this track, each participating team provided point forecasts for the daily sales of 30,490 Walmart products over a span of 28 days. These individual forecasts were then summed to represent forecasts of more aggregate quantities, such as forecasts of the sales of a product type (department and category) or a selling location (store and state), resulting in a total of 12 aggregation levels. Of the 12 aggregation levels, we use the forecasts from level 2 to level 9 (see a brief description in Table 4). For example, level 2, which is aggregated by state, consists of unit sales of three U.S. states: California, Wisconsin, and Texas.[3] For more details of the M5 competition data set, we refer interested readers to Makridakis et al. (2022a).

5.1.2. M4 Data Description. The M4 competition covers different domains (micro, industry, macro, finance, demographic, and other) and different frequencies (yearly, quarterly, monthly, weekly, daily, and hourly). In our empirical analysis, we focus on the hourly frequency data, where participants were asked to provide point forecasts for unknown variables across 48 hours in the future (for other frequencies, the number of data points to be forecast is fewer than 20). In this way, the sample size of 48 is larger in the M4 data set, complementing the sample size of 28 in the M5 data set.

As mentioned, for the pooled inference to be considered as a viable candidate, the forecast errors across different variables need to be dependent. However, the hourly frequency data may have different starting times. To improve the chance of having more dependent series, we focus on the 119 series with the same starting time (noon on July 1st, 2015). (In other words, there are a total of 119 variables.) In line with the M5 competition data set, in our analysis, we only use submissions from the top 17 participants that beat the theta benchmark (Makridakis et al. 2020). For more details on the M4 competition data set, we refer interested readers to Makridakis et al. (2020).

Table 4. Summary of the M5 Data Set Used in the Empirical Analysis

Level	No. of variables	Description
2	3	Unit sales aggregated by state
3	10	Unit sales aggregated by store
4	3	Unit sales aggregated by category
5	7	Unit sales aggregated by department
6	9	Unit sales aggregated by state and category
7	21	Unit sales aggregated by state and department
8	30	Unit sales aggregated by store and category
9	70	Unit sales aggregated by store and department

5.2. Benchmark Aggregation Methods

Recall from Section 4 that our proposed pooled inference method is denoted by P:Linear. Next, we introduce various benchmark aggregation methods that are based on separate inference (we use S as the initial letter in the following acronyms to indicate separate inference).

1. S:EW. Equally weighted aggregation or simple average.
2. S:Sample. The vanilla sample covariance estimator implemented using the posterior mean under an unknown covariance matrix in Winkler (1981).
3. S:Linear. The linear shrinkage covariance estimator (Ledoit and Wolf 2004).
4. S:Cor. The common correlation estimator, which treats the pair-wise correlation between any two experts as constant (Soule et al. 2024). To estimate the common correlation, we use the Bayesian maximum posterior estimator with an uninformative prior.
5. S:S + EW. Optimally combine the aggregated forecasts from two methods: S:Sample and S:EW (Blanc and Setzer 2020).

Table 5. Performance Summary of Different Methods Where Each Method Aggregates Forecasts from the Top Two Teams in the M5 Competition

Level	No. of variables	Pooled inference P:Linear	Separate inference S:EW	S:Sample	S:Linear	S:Cor	S:S + EW	S:Var	S:Rob
2	3	0.301	**0.296**	0.302	0.300	0.300	0.298	0.300	0.307
3	10	0.378	0.402	0.388	0.378	0.378	0.386	**0.377**	0.404
4	3	**0.245**	0.256	0.252	0.245	0.246	0.249	0.249	0.263
5	7	0.333	0.352	**0.330**	0.331	0.331	0.331	0.333	0.345
6	9	**0.363**	0.378	0.370	0.366	0.366	0.365	0.367	0.398
7	21	**0.449**	0.471	0.454	0.449	0.450	0.451	0.451	0.474
8	30	**0.448**	0.480	0.471	0.454	0.454	0.462	0.453	0.493
9	70	**0.542**	0.574	0.559	0.546	0.546	0.551	0.546	0.596

Notes. The number of samples used for covariance matrix estimation is 27. In line with the competition evaluation metric, the performance is measured in weighted root mean squared scaled error. The best-performing method for each scenario is in bold. The top teams are chosen based on their overall performance in the M5 competition.

6. S:Var. The variance weights estimator, which is similar to S:Cor with the pair-wise correlation between any two experts set to zero (Bunn 1985, Schmittlein et al. 1990).

7. S:Rob. The robust covariance estimator, which accounts for potential outliers in covariance estimation (Rousseeuw and Leroy 1987, Rousseeuw and Van Zomeren 1990, Rousseeuw and Driessen 1999).

5.3. Empirical Procedures

For both M5 and M4 data sets, we consider aggregating the forecasts provided by $k = 2, 3, 10, 15$ teams, with two alternative ways of selecting the k teams: the top k teams of the competition or k heterogeneous teams selected by solving the maximum diversity problem. We adopt leave-one-out procedures to evaluate the performance of different methods. More specifically, in the M5 (M4) data sets, we use 27 (47) samples to estimate the covariance matrix, which serves as inputs to make an out-of-sample prediction for the left-out data point. This procedure is repeated to obtain forecasts for all 28 (48) data points.

It is worth noting that in the M5 setting, the pooled inference uses forecasts of all of the variables on the same aggregation level (the second column of Table 4 displays the number of variables on each aggregation level). However, the M4 data set lacks a hierarchical structure. To emulate the variable pooling scenarios considered in the M5 analysis, we randomly choose $m = 3, 7, 9, 10, 21, 30, 70$ number of variables from the total of 119 variables in the M4 data set, and for each given m, this process is iterated 100 times.

5.4. Empirical Results

To simplify the presentation of empirical findings, in the main text, we present the results of the aggregation problem where each variable is forecast by the top 2 or top 15 teams of the competition. The results remain robust to the alternative crowd sizes considered (top 3 or top 10 teams) or to the alternative choice of more heterogeneous k teams. Hence, these results are presented in Online Appendix F as robustness checks. In the following, we present the results from the M5 data set first followed by those from the M4 data set.

5.4.1. M5 Results. The main empirical results using the M5 data set are presented in Tables 5 and 6, which summarize the performances of different methods

Table 6. Performance Summary of Different Methods Where Each Method Aggregates Forecasts from the Top 15 Teams in the M5 Competition

Level	No. of variables	Pooled inference P:Linear	Separate inference S:EW	S:Sample	S:Linear	S:Cor	S:S + EW	S:Var	S:Rob
2	3	0.306	**0.281**	0.493	0.283	0.293	0.331	0.288	1.239
3	10	0.404	0.374	0.618	0.369	0.363	0.374	**0.356**	1.076
4	3	0.240	0.253	0.286	0.228	0.234	**0.228**	0.246	0.768
5	7	0.310	0.350	0.460	**0.290**	0.293	0.323	0.323	0.762
6	9	0.363	0.370	0.557	**0.343**	0.356	0.372	0.360	1.137
7	21	0.431	0.459	0.598	**0.402**	0.420	0.437	0.440	1.208
8	30	0.446	0.459	0.666	**0.437**	0.437	0.454	0.437	1.252
9	70	0.537	0.558	0.810	**0.524**	0.526	0.558	0.534	1.414

Notes. The number of samples used for covariance matrix estimation is 27. In line with the competition evaluation metric, the performance is measured in weighted root mean squared scaled error. The best-performing method for each scenario is in bold. The top teams are chosen based on their overall performance in the M5 competition.

Table 7. Performance Summary of Different Methods Where Each Method Aggregates Forecasts from the Top Two Teams in the M4 Competition

No. of variables	Pooled inference P:Linear	Separate inference						
		S:EW	S:Sample	S:Linear	S:Cor	S:S + EW	S:Var	S:Rob
3	**0.747** (86%)	1.103	0.816	0.819	0.819	0.814	0.845	0.899
7	**0.661** (93%)	1.091	0.809	0.812	0.813	0.806	0.837	0.894
9	**0.644** (90%)	1.129	0.822	0.828	0.827	0.821	0.851	0.903
10	**0.617** (93%)	1.095	0.806	0.810	0.809	0.804	0.832	0.888
21	**0.532** (100%)	1.098	0.803	0.807	0.807	0.802	0.829	0.886
30	**0.471** (100%)	1.100	0.799	0.804	0.804	0.798	0.828	0.884
70	**0.378** (100%)	1.097	0.797	0.802	0.802	0.796	0.826	0.883

Notes. The number of samples used for covariance matrix estimation is 47. The numbers reported are the average root mean squared scaled error across 100 iterations. Moreover, under P:Linear, the numbers in the parentheses represent the percentage of times (of 100 iterations) that P:Linear is the best-performing method relative to the benchmarks based on separate inference. The best-performing method for each scenario is in bold. The top teams are chosen based on their overall performance in the M4 competition.

across various aggregation levels when aggregating forecasts of the top 2 and top 15 teams, respectively. In Table 5, although P:Linear achieves the best performance in five of the eight levels (ranging from levels 2 to 9), its advantage over the benchmarks is marginal. In Table 6, P:Linear never emerges as the top-performing method. Overall, P:Linear does not show superior performances in the M5 competition data set.

To interpret these results, let us focus on the separate inference methods. It is helpful to recognize that S:EW performs competitively compared with other methods that apply differential weighting. Notably, the performance gap between S:EW and the best-performing separate inference method is only at most 18%. The strong performance of S:EW provides insight into two key reasons why P:Linear does not demonstrate superior performance in the M5 competition data set. First, the strong performance of S:EW indicates low expert heterogeneity. This is consistent with our anecdotal evidence that all top teams utilize the LightGBM method in the M5 competition (see Table 3). When expert heterogeneity is low, the pooled inference offers little benefit (Proposition 2); therefore, P:Linear is not expected to demonstrate superior performance. Second, the strong performance of S:EW indicates that the forecast errors may not be stationary across time. This can introduce bias in the estimation as the training data used for covariance estimation may not accurately reflect the true covariance structure of the test data. As a result, this bias amplifies estimation errors, especially in the pooled inference because of the large covariance matrix, which adversely impacts the performance of P:Linear. As robustness checks, we observe similar findings using an alternative way of selecting teams in the M5 data set (see Tables EC.1, EC.2, EC.5, EC.6, EC.9, and EC.10 in Online Appendix F).

5.4.2. M4 Result. Recall that the M4 data set is not hierarchical; therefore, the variables to be considered for pooling are not as straightforward as those in the M5 setting. Instead, in each scenario, we randomly choose the variables from the total of 119 variables, and we do so 100 times for each given number of variables.

The main empirical results using the M4 data set are presented in Tables 7 and 8 when aggregating forecasts of the top 2 and top 15 teams, respectively. Overall, P:Linear consistently outperforms the benchmark methods in contrast to its relatively mediocre

Table 8. Performance Summary of Different Methods Where Each Method Aggregates Forecasts from the Top 15 Teams in the M4 Competition

No. of variables	Pooled inference P:Linear	Separate inference						
		S:EW	S:Sample	S:Linear	S:Cor	S:S + EW	S:Var	S:Rob
3	**0.437** (57%)	1.086	0.463	0.546	0.683	0.439	0.784	0.677
7	**0.384** (88%)	1.081	0.467	0.547	0.682	0.441	0.780	0.682
9	**0.374** (87%)	1.109	0.464	0.537	0.676	0.434	0.783	0.651
10	**0.361** (91%)	1.081	0.460	0.544	0.680	0.435	0.769	0.657
21	**0.326** (99%)	1.070	0.461	0.540	0.674	0.434	0.766	0.653
30	**0.298** (100%)	1.081	0.450	0.530	0.666	0.427	0.763	0.648
70	**0.276** (100%)	1.079	0.445	0.529	0.665	0.423	0.762	0.642

Notes. The number of samples used for covariance matrix estimation is 47. The numbers reported are the average root mean squared scaled error across 100 iterations. Moreover, under P:Linear, the number in the parentheses represents the percentage of times (of 100 iterations) that P:Linear is the best-performing method relative to the benchmarks based on separate inference. The best-performing method for each scenario is in bold. The top teams are chosen based on their overall performance in the M4 competition.

performance in the M5 competition data set. Notably, across all scenarios considered, P:Linear is the best-performing method with the lowest average root mean squared scaled error (RMSSE). Besides, for each scenario, of the 100 iterations, there is a sizeable proportion of times (in the range between 64% and100%) that P:Linear achieves the lowest RMSSE relative to the benchmark methods considered.

To understand the superior performance of P:Linear in the M4 competition data set, we can again start with S:EW. In contrast to the M5 results in Tables 5 and 6, S:EW performs poorly relative to other separate inference methods in Tables 7 and 8, with the performance gap between S:EW and the best-performing separate inference method as large as 60%. Following the discussion of the M5 results, the weak performance of S:EW indicates two key characteristics of the M4 competition data set. (i) Expert heterogeneity is high in the M4 competition, and (ii) forecast errors tend to be stationary—both of which contrast with the characteristics of the M5 competition data set. Factor (i) is also consistent with our anecdotal evidence that the top teams in the M4 competition utilize a diverse set of methods (see Table 3). Taken together, factors (i) and (ii) create a favorable setting for the pooled inference as (i) highlights the value of applying differential weighting and (ii) suggests that the data allow for reliable estimation of these weights. This explains why P:Linear, which can estimate the weights more effectively, performs well in this setting.

Another intriguing observation shown in Tables 7 and 8 is that the performance of P:Linear improves as we pool more variables.[4] This is manifested in two different ways. (i) The average performance increases, and (ii) the percentage of times that P:Linear is the best-performing method also shows an increasing trend. Intuitively, as we consider more variables in the pooled inference, there is a larger informational gain because of higher model accuracy (i.e., lower bias). Meanwhile, for a fixed sample size, the estimation burden also tends to increase because there are more parameters to be estimated, potentially leading to higher estimation error (i.e., higher variance). To deal with this high-variance issue, the linear shrinkage covariance estimator in P:Linear tends to apply a stronger shrinkage level when estimating the covariance matrix (Ledoit and Wolf 2004, section 4.3.1). More specifically, the linear shrinkage covariance estimator tries to strike a balance between the bias and variance of the covariance estimator (Ledoit and Wolf 2004, equation 10). For a fixed number of pooled variables, the estimator approximately achieves the optimal bias-variance trade-off (Ledoit and Wolf 2004, theorem 3.2). Comparing performances of P:Linear with different numbers of variables pooled can be viewed as comparing the performances across different bias-variance trade-off curves. Our empirical findings in Tables 7 and 8 reveal that as we pool more variables, the performances of P:Linear across different bias-variance trade-off curves tend to improve. We also observe similar findings using an alternative way of selecting teams in the M4 data set (see Tables EC.3, EC.4, EC.7, and EC.8 in Online Appendix F).

On a high level, this empirical result seems at odds with the well-known "forecast combination puzzle" (Stock and Watson 2004, p. 428), which finds that combination using equal weights often empirically dominates more sophisticated weighting schemes, such as the optimal weights. This puzzle is based on comparing performances of different aggregation methods on a specific bias-variance curve. In contrast, our empirical findings, which indicate that the pooled inference performance improves in the number of variables pooled, stem from comparing P:Linear across different bias-variance curves.

5.5. Additional Analysis Using Simulation

As discussed, a surprising finding from the M4 results is that the performance of the pooled inference improves as the number of pooled variables increases. To assess the robustness of this result, we conduct additional simulations.

5.5.1. Simulation Setup and Procedures. Specifically, we aggregate forecasts from 15 experts, with the possibility of pooling $k = 3, 7, 9, 10, 21, 30, 70, 700, 7{,}000$ variables. Notably, because the largest k considered in our empirical setting is 70, the cases of $k = 700$ and $k = 7{,}000$ allow us to examine the scalability of our pooled method in estimating substantially larger covariance matrices. For instance, pooling $k = 7{,}000$ variables when aggregating forecasts of 15 experts requires estimating a $105{,}000 \times 105{,}000$ covariance matrix.

Our simulation setup follows the data-generating process in (1) with $c_i = 1$ and $Var(z) = 0$: that is, $u_{i,j} = x_i + y_j + \epsilon_{i,j}$. We set $Var(x_i) = 1$ and $Var(\epsilon_{i,j}) = 30$. We consider two settings that correspond to low and high expert heterogeneity by choosing different values for $Var(y_j)$. To align with the M4 setting, we simulate 47 observations for each $u_{i,j}$, where i depends on the number of variables pooled and $j = 1, 2, \ldots, 15$. These observations would serve as historical data to estimate the covariance matrix and perform the pooled inference using P:Linear. For the alternative separate inference methods, we consider equally weighted aggregation (S:EW) and the optimal performance (S:Opt), where S:Opt represents the best-possible (optimal) performance that the separate inference can achieve. For each of the two simulation settings, we simulate 100 iterations. The average performances of the pooled inference (P:Linear) and the percentage of time that it outperforms alternative methods are reported below. Both

S:EW and S:Opt are computed based on theoretical formulas. Hence, they remain the same regardless of the number of pooled variables.

5.5.2. Handling Computational Burden. To handle the operations of large covariance matrices under the pooled inference, we implement the linear shrinkage covariance estimator in a more efficient way (instead of using the linshrink_cov function from the nlshrink package). This new implementation reformulates the optimal pooled weights without requiring matrix inversion as presented in Equation (9). Specifically, we apply singular value decomposition to avoid the storage and computational burden of multiplying two large matrices when computing the optimal shrinkage parameter. Additionally, we express the optimal pooled weights in terms of singular values and their corresponding singular vectors, eliminating the need to invert a large covariance matrix. All of the simulation runs are conducted on a MacBook Air with an Apple M3 chip and 16 GB of memory.

5.5.3. Simulation Results. Panel A of Table 9 shows the simulation performance of different methods when experts exhibit low heterogeneity (the low expert heterogeneity is also reflected in the small gap between S:EW and S:Opt). Overall, the average performances of P:Linear are poor, and not a single iteration outperforms S:EW or S:Opt. This aligns with our theoretical prediction as well as the M5 results. In contrast, panel B of Table 9 shows the results when experts exhibit high heterogeneity (the high expert heterogeneity is also reflected in the large gap between S:EW and S:Opt). Overall, under P:Linear, the average root mean squared scaled errors improve as more and more variables are pooled. Similarly, the percentage of time that P:Linear is the best-performing method also improves as more and more variables are pooled. For example, when the number of variables pooled is three, P:Linear is never the top-performing method in all of the 100 simulations. In contrast, when the number of variables pooled increases to 7,000, P:Linear achieves the lowest average RMSSE, and it is the best-performing method in 100% of 100 simulations.

Beyond evaluating the performance of P:Linear, it is also important to assess its computational burden, particularly because P:Linear often requires estimating significantly larger covariance matrices compared with separate inference. Table 10 reports the average computation time for a single simulation run of P:Linear based on 100 simulations for both low and high expert heterogeneity. Overall, as the number of pooled variables increases from 3 to 7,000, computation time rises considerably. However, even in the most computationally intensive scenario of pooling 7,000 variables in our simulation, P:Linear successfully estimates a $105{,}000 \times 105{,}000$ covariance matrix within just 90 seconds. This suggests that the additional computational cost remains manageable.

In sum, this simulation study not only confirms the superior performances of P:Linear observed in the M4 results but also, highlights the scalability of P:Linear in estimating large covariance matrices within a reasonable computational time.

5.6. Discussion

The aim of this subsection is to further shed light on the practicality of the pooled inference method. So far, we have considered pooling at most $k = 7{,}000$ variables in our simulations. This is because the marginal

Table 9. Performance Summary of Different Methods Where Each Method Aggregates Forecasts from 15 Experts When Experts Exhibit (Panel A) Low Heterogeneity or (Panel B) High Heterogeneity

	Panel A: Low heterogeneity			Panel B: High heterogeneity		
	Pooled inference	Separate inference		Pooled inference	Separate inference	
No. of variables	P:Linear	S:EW	S:Opt	P:Linear	S:EW	S:Opt
3	1.837 (0%)	1.791	**1.789**	2.194 (0%)	2.273	**2.044**
7	1.816 (0%)	1.791	**1.789**	2.146 (0%)	2.273	**2.044**
9	1.811 (0%)	1.791	**1.789**	2.123 (0%)	2.273	**2.044**
10	1.827 (0%)	1.791	**1.789**	2.136 (0%)	2.273	**2.044**
21	1.819 (0%)	1.791	**1.789**	2.073 (19%)	2.273	**2.044**
30	1.829 (0%)	1.791	**1.789**	2.054 (45%)	2.273	**2.044**
70	1.826 (0%)	1.791	**1.789**	**2.005** (92%)	2.273	2.044
700	1.823 (0%)	1.791	**1.789**	**1.969** (100%)	2.273	2.044
7,000	1.816 (0%)	1.791	**1.789**	**1.955** (100%)	2.273	2.044

Notes. The number of samples used for covariance matrix estimation is 47. Under P:Linear, the numbers reported are the average mean squared error across 100 iterations. Moreover, the numbers in the parentheses represent the percentage of times (of 100 iterations) that P:Linear is the top-performing method relative to S:EW (equal weights) and S:Opt (optimal performance under the separate inference). The best-performing method for each scenario is in bold. The parameter values used are $Var(x_i) = 1$, $Var(\epsilon_{ij}) = 30$, and $Var(y_j) = 1$ for $j = 1, 2, \ldots, 7$ and $Var(y_j) = 5$ (low expert heterogeneity) or $Var(y_j) = 60$ (high expert heterogeneity) for $j = 8, 9, \ldots, 15$.

Table 10. Average Computational Time (Seconds) Across for 100 Simulations

Setting	3	7	9	10	21	30	70	700	7,000
Low heterogeneity	<0.01	0.09	0.10	0.11	0.16	0.21	0.44	3.61	83.91
High heterogeneity	<0.01	0.09	0.10	0.10	0.15	0.20	0.37	3.63	85.91

Note. We conduct our computations on a MacBook Air with an Apple M3 chip and 16 GB of memory.

benefit of an even larger k beyond $k = 7{,}000$ seems negligible for P:Linear as shown in Table 9. Of course, considering even larger values of k could naturally result in a heavier computational burden.

To illustrate the practical relevance of the pooled inference, let us take sales forecasting in retail settings as an example. Although the number of stock-keeping units can be in the millions, our prescription is that the decision maker should not blindly pool all of the variables. Instead, a prerequisite for using the pooled inference is that the variables to be pooled are related (i.e., the forecast errors across different variables are correlated). For example, in forecasting sales of a product in a neighborhood, one may consider pooling sales of similar products at close zip codes (e.g., New York City has around 150 distinct zip codes). In this case, the number of potential variables to be pooled might be moderate, for which $k = 7{,}000$ is reasonable. In addition, as suggested by our theory and also validated by the empirical and simulation results, the decision maker should only consider the pooled inference when experts exhibit high heterogeneity.

6. Conclusion

In this paper, we propose a new method of combining forecasts from multiple variables and experts when drawing inference about an unknown variable. Extensive work in the literature has primarily focused on aggregating forecasts from multiple experts concerning a single unknown variable into a consensus forecast (referred to as the separate inference). However, in practice, each expert frequently provides a forecast for each variable across multiple variables. Because the forecast errors tend to be correlated across variables, to infer an unknown variable, the decision maker may find it beneficial to pool forecasts from other variables (referred to as the pooled inference). In general, we show that the value of additional information from pooling is nonnegative. Interestingly, we find two cases where the value of additional information from pooling is zero. (i) Experts are exchangeable (no need to weight the forecasts from different experts differently), and (ii) there is no idiosyncratic noise (no further room to reduce the idiosyncratic noise). Hence, a key managerial implication is that the decision maker should only consider the pooled inference when experts are substantially heterogeneous. Lastly, we illustrate how the pooled inference can be implemented and demonstrate that it can outperform benchmarks based on separate inference in some forecasting environments. Additionally, even when pooling a large number of variables—necessitating the handling of high-dimensional covariance matrices—pooled inference remains computationally manageable.

There are several future directions in which our paper could be extended. First, given the promising performance of the pooled inference in point forecast aggregation, one can apply the notion of pooling to probability forecasts because it is common for experts to forecast multiple variables in probabilistic forecasting (Winkler et al. 2019). This extension is not straightforward because of many differences between point and probability forecasts. For example, different types of forecasts are evaluated using different proper scoring rules, which may incentivize different forecasting behaviors. As a result, different types of forecasts may require different mathematical models. As a second direction for future research, the concept of pooling is not limited to different variables. For example, experts may provide multiple probabilistic forecasts, such as multiple quantiles for a given variable (Bansal et al. 2017, Makridakis et al. 2022b). In this case, the decision maker can pool forecasts provided for the other quantiles to infer about a particular quantile. This is another direction to explore.

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Endnotes

[1] From a modeling perspective, based on the data-generating process in Satopää (2021) (specifically, Satopää 2021, equation 4, p. 1733), it is easy to verify that the forecast errors are independent across variables. Hence, the drivers of "borrowing strength" in the two papers differ.

[2] If the variable-specific randomness has different variances such that $u_{i,j} = \tilde{\sigma}_i(x_i + c_i(y_j + z) + \epsilon_{i,j})$, we can redefine the new forecast

errors to be $u'_{i,j} = u_{i,j}/\tilde{\sigma}_i$, and then, the new data-generating process $u'_{i,j} = x_i + c_i(y_j + z) + \epsilon_{i,j}$ will be equivalent to (1). Furthermore, here we assume that x_i, $1 \le i \le m$, are independent; in Online Appendix C, we show that our key theoretical insights are generalizable to dependent settings (see the discussion in Remark EC.1 in Online Appendix C).

[3] We do not consider level 1 because there is only one variable (grand total) at that level. Moreover, we do not consider levels 10, 11, and 12 because daily sales on these three levels tend to be intermittent, often with most of the days having zero sales and some days having single-digit sales.

[4] Two remarks are useful to understand related empirical results. First, for each given separate inference method, the performances are similar across different rows in Tables 7 and 8. Recall that for each scenario, we randomly pick the variables each time (for a total of 100 times). Because each variable is equally likely to be selected, the average performance of any given separate inference method is expected to be invariant with respect to the number of variables pooled. Second, unlike the M4 analysis, the performances are not comparable across different aggregation levels in our previous M5 analysis because the variables in each aggregation level are different.

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